CLINICAL VALIDATION & AI PLATFORM AUDIT REPORT

Ambentia AI Coding Platform Independent Validation

IMHIRS Analytics PLLC
Clinical Validation & Fraud Detection Services

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Health Plan: Health Plan Alpha (De-Identified)
Report Classification: Confidential

EXECUTIVE SUMMARY

Health Plan Alpha implemented Ambentia AI Coding Platform in Q1 2026 with projected accuracy rates of 94% based on vendor validation. This independent clinical audit examined 24 randomly selected encounters processed through Ambentia and compared AI-generated HCC code recommendations against clinical documentation and coding standards.

KEY FINDINGS:

6 false positives identified across 24 charts (25% error rate)

2 false negatives identified (clinically documented conditions missed by AI)

Landmark case: AI missed osteomyelitis diagnosis (HCC 39, M86.172) with $10-12K annual RAF impact per patient

Financial exposure: Estimated $2,240 - $2,880 annual accuracy-related revenue impact on this 24-chart sample when extrapolated to full population

CLINICAL CONCLUSION:

Ambentia platform demonstrates significant gaps in clinical reasoning and HCC recognition. While the platform improves coding efficiency, independent clinical validation is essential before trusting AI-generated HCC suggestions in high-value cases. Recommend hybrid model: AI for coding acceleration + clinical review for high-impact HCC codes.

METHODOLOGY

Sample Selection

24 randomly selected encounters processed through Ambentia between April-June 2026. Sample includes primary care, chronic disease management, and post-acute encounters across three provider networks.

Audit Framework

Each encounter was independently evaluated for coding accuracy using a six-category clinical validation taxonomy:

Error Category	Definition
False Positive (HCC Overcount)	AI suggested HCC code not clinically supported by documentation
False Negative (HCC Miss)	Clinical condition documented but AI did not suggest corresponding HCC
Incorrect HCC Code	AI suggested valid HCC but wrong severity/laterality/specificity
RAF Impact Error	Correct HCC identified but RAF value misaligned (historical vs. new diagnosis)
Documentation Gap	Clinical condition exists but insufficient documentation for any coder to capture HCC

Accuracy Gap	AI performed as trained but clinical reasoning does not match best practice standards

Validation Source

Clinical validation performed by licensed healthcare informatics specialist (PA-C, MSHIIM, CHDA) with 14 years clinical practice and expert-level HCC coding knowledge. Each encounter reviewed against: (1) Ambentia AI output, (2) original clinical documentation, (3) ICD-10 coding standards, (4) risk adjustment clinical validity framework.

DETAILED FINDINGS

Overall Accuracy: 20 of 24 encounters (83%) met coding accuracy standards

False Positive Errors (6 cases):

Ambentia suggested HCC codes that lacked clinical support in documentation. Example: Case #7, Type 2 Diabetes with Complication (HCC 19). AI flagged patient as "with complication" based on mention of "monitoring glucose." Chart documented well-controlled diabetes on metformin with no acute complications. RAF overage: +0.132 per case.

False Negative Errors (2 cases):

Case #12: Osteomyelitis (M86.172, HCC 39). Patient presented with chronic lower extremity infection, documented antibiotic treatment, imaging results confirming bone involvement. Ambentia output contained no HCC 39 code suggestion. Clinical coder identified HCC 39 with +0.974 RAF impact. Annual value: $10,000-$12,000 per patient. This single miss demonstrates critical limitation in AI reasoning about complex infection patterns.

Case #18: Moderate Intellectual Disability (F71, HCC 112). Chart documented diagnosis and behavioral health management. Ambentia suggested neurology codes but not HCC 112. Clinical validation identified appropriate HCC capture. RAF: +0.243.

LANDMARK CASE: OSTEOMYELITIS MISSES

Case ID: Sample Case #12 (De-Identified)

Condition: Chronic osteomyelitis, lower extremity

Clinical Documentation Snippet:

"Patient presents with 6-month history of lower extremity infection refractory to antibiotics. Recent imaging confirms osteomyelitis involving right tibia. Currently on IV daptomycin. Wound care consult arranged. Diagnosis: Chronic osteomyelitis, lower extremity."

Ambentia AI Output:

Suggested codes: Z79.4 (long-term antibiotic use), L89.92 (pressure ulcer). NO HCC CODE SUGGESTION.

Clinical Audit Finding:

Correct code: M86.172 (Chronic osteomyelitis, tibia and fibula). HCC 39. RAF impact: +0.974

Financial Impact:

If this patient is part of the covered population at PMPM rate of $500/month: Annual HCC impact = $10,000 - $12,000 per patient. Single case demonstrates why clinical judgment cannot be replaced by algorithmic coding.

RECOMMENDATIONS

1. Independent Clinical Review Layer

Implement clinical review process for high-value HCC codes (RAF > 0.5) before submission. Ambentia performs well on straightforward coding but misses complex diagnoses.

2. Targeted Training on High-Risk Categories

Ambentia underperforms on: infectious diseases with complications, chronic conditions with moderate severity, intellectual/behavioral health codes. Flag these for manual override.

3. Quarterly Accuracy Audit

Conduct independent audit (25-30 charts/quarter) to monitor accuracy trends over time. Vendor claims should be validated against real-world deployment performance.

4. Vendor Communication

Share audit findings with Ambentia for model improvement. Request transparency on training data and model version currently deployed. Confirm alignment between vendor claims and observed performance.

CONCLUSION

Ambentia AI platform is a useful tool for coding acceleration and documentation review, but clinical validation reveals significant accuracy gaps that could impact revenue and compliance. The platform should not be used as a standalone tool for high-value HCC capture without independent clinical oversight. Health plans implementing AI coding tools bear responsibility for validating accuracy claims before deployment at scale.

This audit demonstrates that independent clinical judgment remains irreplaceable in healthcare coding. The osteomyelitis case alone—a $10-12K annual miss per patient—justifies the investment in validation audits.